

Lab Exercise 2.2: Getting to know Cisco Firepower/ASA Firewall**Learning Outcomes**

Upon completion of this laboratory exercise, you should be able to:

- Access Firepower/ASA firewall via console port
- Configure security-level for segmenting networks into security zones
- Configure ACL for overriding default security policy
- Configure Modular Policy Framework (MPF) for other security features
- Initialize and reload Firepower/ASA firewall

Required Hardware

- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

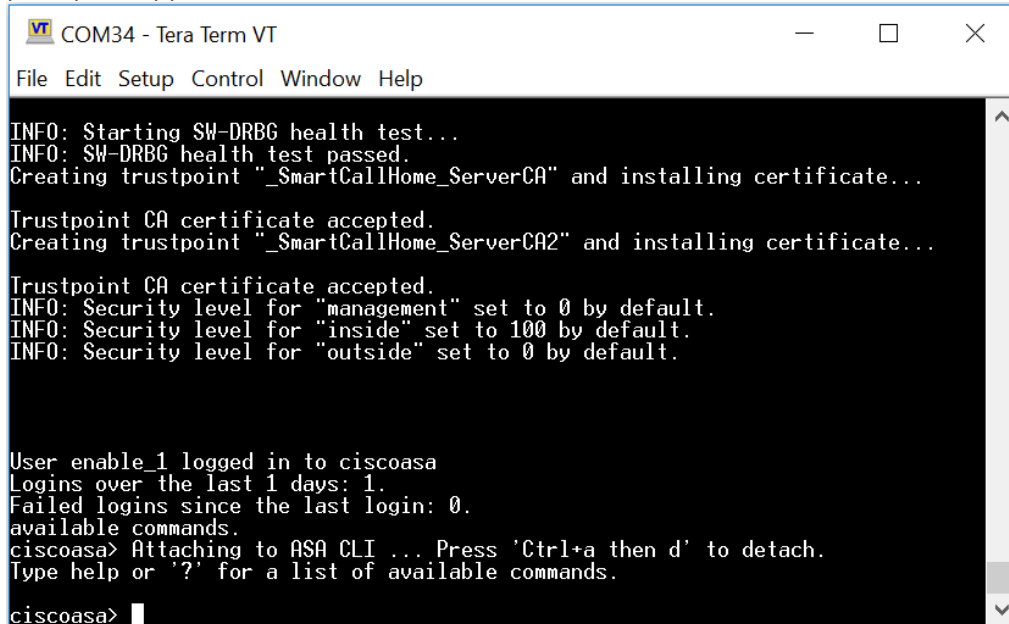
PART 1: Accessing Cisco Firepower/ASA Firewall via Console Port

- 1.1 Similar as Cisco switches and routers, Cisco Firepower/ASA firewall can be configured using CLI commands. However, note that some commands are slightly different.
- 1.2 To begin, power up Cisco Firepower/ASA and connect a console cable from your PC to the serial console port as shown in the figure below:



Console Port

- 1.3 Start Tera Term, press the 'Enter' key if necessary and wait for the user EXEC command prompt to appear as shown:



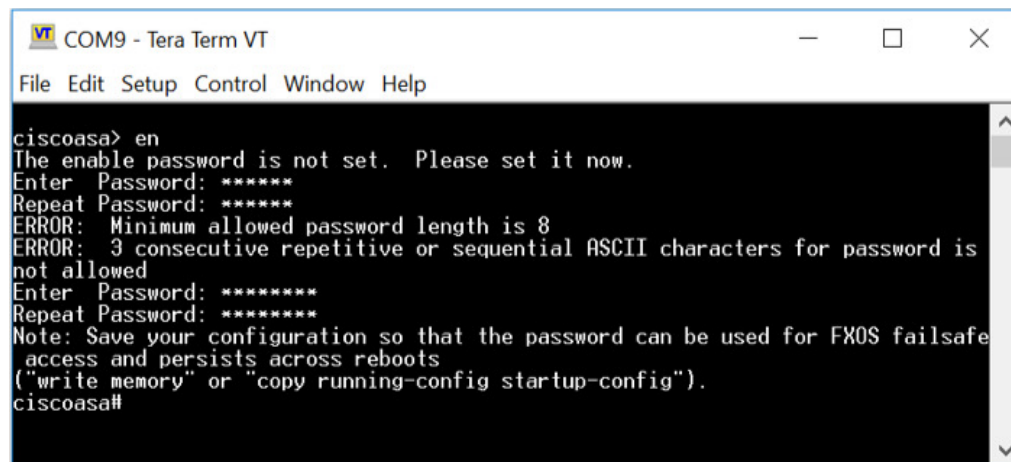
```
COM34 - Tera Term VT
File Edit Setup Control Window Help

INFO: Starting SW-DRBG health test...
INFO: SW-DRBG health test passed.
Creating trustpoint "_SmartCallHome_ServerCA" and installing certificate...
Trustpoint CA certificate accepted.
Creating trustpoint "_SmartCallHome_ServerCA2" and installing certificate...
Trustpoint CA certificate accepted.
INFO: Security level for "management" set to 0 by default.
INFO: Security level for "inside" set to 100 by default.
INFO: Security level for "outside" set to 0 by default.

User enable_1 logged in to ciscoasa
Logins over the last 1 days: 1.
Failed logins since the last login: 0.
available commands.
ciscoasa> Attaching to ASA CLI ... Press 'Ctrl+a then d' to detach.
Type help or '?' for a list of available commands.
ciscoasa> |
```

- 1.6 To go to the privileged EXEC mode, enter **enable** command similar as Cisco switches and routers. Note that **NO** enable password is set but there is a need to enter new password to proceed. For consistency and ease of remembering, please enter:

Enter Password: **Student@s1t**

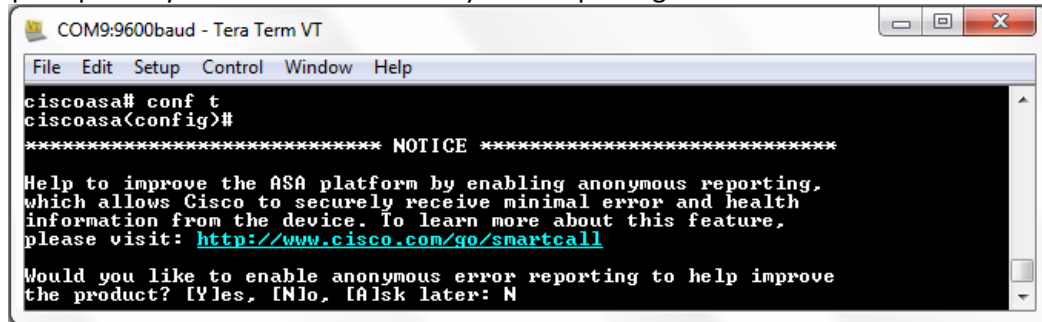


```
COM9 - Tera Term VT
File Edit Setup Control Window Help

ciscoasa> en
The enable password is not set. Please set it now.
Enter Password: *****
Repeat Password: *****
ERROR: Minimum allowed password length is 8
ERROR: 3 consecutive repetitive or sequential ASCII characters for password is
not allowed
Enter Password: *****
Repeat Password: *****
Note: Save your configuration so that the password can be used for FXOS failsafe
access and persists across reboots
("write memory" or "copy running-config startup-config").
ciscoasa#
```

Warning: Let's be caring and responsible. Please do NOT enter "write memory" or "copy running-config startup-config" to save your password onto any network device without being told to do so as this will prevent others from using if you happen to enter a different password due to typo errors. Violators will have marks deducted.

- 1.7 To go to the global configuration mode, enter the same **config t** command. When prompted if you wish to enable anonymous reporting as shown below. Answer 'N'.



```

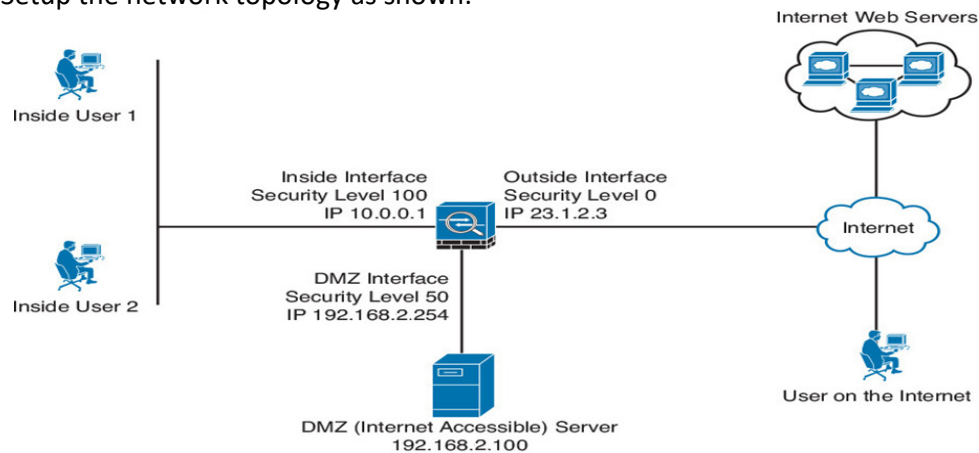
COM9:9600baud - Tera Term VT
File Edit Setup Control Window Help
ciscoasa# conf t
ciscoasa(config)#
***** NOTICE *****
Help to improve the ASA platform by enabling anonymous reporting,
which allows Cisco to securely receive minimal error and health
information from the device. To learn more about this feature,
please visit: http://www.cisco.com/go/smartcall
Would you like to enable anonymous error reporting to help improve
the product? [Y]es, [N]o, [A]sk later: N

```

- 1.8 You are now ready to start configuring the Firepower/ASA firewall.

PART 2: Configuring Security-Level for Segmenting Networks into Security Zones

- 2.1 Setup the network topology as shown:



- 2.2 Connect 1st PC simulating **User on the Internet** to interface **E1/1** of the Firepower/ASA and name it **Outside** with default security-level 0, 2nd PC simulating **DMZ Server** to interface **E1/3** and name it **DMZ** with security-level, say 50, and 3rd PC simulating **Inside User** to interface **E1/5** and name it **Inside** with default security-level 100.

Notes: By default, interface **Outside** is pre-configured with dynamic NAT. To remove NAT, configure:

```

ciscoasa(config)# object network obj_any
ciscoasa(config-network-object)# no nat (any,outside) dynamic interface

```

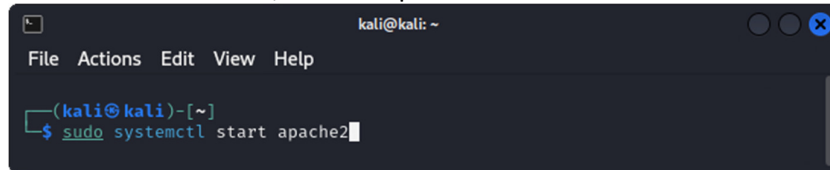
In addition, name **Inside** is already pre-configured for int vlan 1 in the Firepower/ASA. To use the same name, you will need to remove it from int vlan 1 by configuring:

```

ciscoasa(config)# int vlan 1
ciscoasa(config-if)# no nameif inside

```

- 2.3 Configure the interfaces of the Firepower/ASA with suitable IP addresses and subnet masks.
- 2.4 When done, your Firepower/ASA should be operational.
- 2.5 Boot up the DMZ Server in Kali Linux, and the others may be in Windows.
- 2.6 Configure suitable IP addresses, subnet masks and default gateways to the PCs.
- 2.7 At the DMZ Server PC, run the apache web server with default web page as shown.



```
kali@kali: ~  
File Actions Edit View Help  
~(kali@kali)-[~]  
$ sudo systemctl start apache2
```

- 2.8 Start a web browser at the Outside PC. Can it access the website hosted at the DMZ Server? Why or why not?
- 2.9 Similarly, start a web browser at the Inside PC. Can it access the website hosted at the DMZ Server? Why or why not?
- 2.10 At the Firepower/ASA, show the state table using the following command when you access the website again in Step 2.9:

```
ciscoasa# sh conn
```

What do you observe?

PART 3: Configuring ACL for Overriding Default Security Policy

- 3.1 By design, DMZ (DeMilitarized Zone) is for an organization to host servers, e.g. web servers, email servers, DNS, etc., for users from the Internet to access.
- 3.2 However by default, Firepower/ASA firewall will block Internet users with lower security level from initiating a connection to the DMZ and inside networks with higher security levels.
- 3.3 Nevertheless, you are able to override the security level access control in the Firepower/ASA using ACLs.

- 3.4 Configure suitable ACL to allow Outside PC to access the website hosted at the DMZ PC.
- 3.5 Now, can the Outside PC access the website hosted at the DMZ Server?
- 3.6 At the Firepower/ASA, show the state table when you access the website again in Step 3.5. What do you observe?

PART 4: Configuring MPF for Other Security Features

- 4.1 For security reason, Firepower/ASA will deny ICMP ping packets by default, which explains why till now you've not been asked to use ping to verify network connectivity.
- 4.2 In Part 4 of this lab exercise, you are going to practice modifying the default MPF configuration to allow ping packets to pass through as discussed.
- 4.3 When done, verify that the Inside PC can indeed ping DMZ PC and Outside PC.
- 4.4 Similarly, show the state table at the Firepower/ASA when you ping again in Step 4.3. What do you observe?

PART 5: Initialising and Reloading Firepower/ASA Firewall

- 5.1 To reset the Firepower/ASA firewall, you may restore it to its factory default settings by entering the commands:

```
ciscoasa# configure t  
ciscoasa(config)# configure factory-default
```

```
WARNING: The boot system configuration will be cleared. The first  
image found in disk0:/ will be used to boot the system on the next  
reload.
```

```
Verify there is a valid image on disk0:/ or the system will not  
boot.
```

```
Begin to apply factory-default configuration:
```

```
Clear all configuration
```

```
...
```

```
<output omitted>
```

5.2 Alternatively, you may erase its startup-config from the flash memory:

```
ciscoasa# write erase

Erase configuration in flash memory? [confirm]
[OK]
ciscoasa#
```

5.3 And then reload it. Remember to respond with **N** if prompted that the config has been modified and needs to be saved.

```
ciscoasa# reload

System config has been modified. Save? [Y]es / [N]o: N
Proceed with reload? [confirm]

ciscoasa#
***
*** --- START GRACEFUL SHUTDOWN ---
Shutting down isakmp
Shutting down File system
***
*** --- SHUTDOWN NOW ---
Process shutdown finished Rebooting.....
CISCO SYSTEMS
...
<output omitted>
```